

Img-Crypt v2 Documentation

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Manish Dalwani

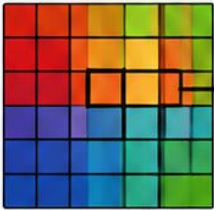
1: Introduction

Img-Crypt is an open-source command-line steganography utility developed in Python. It enables users to hide (encode) and retrieve (decode) secret text messages inside image files using Least Significant Bit (LSB) steganography. The tool is designed with simplicity in mind, making it suitable for students, cybersecurity learners, CTF participants, & digital forensics enthusiasts.

Unlike many traditional steganography tools that rely entirely on command-line arguments, Img-Crypt provides an interactive menu-driven interface while still supporting encryption of hidden messages.

WHAT IS LSB (LEAST SIGNIFICANT BIT) STEGANOGRAPHY?

A digital image is made up of pixels.
Each pixel has 3 colors – Red, Green, Blue (RGB).
Each color is stored in 8 bits (1 byte).



One Pixel

R 10110110

G 01101101

B 11001011

What is LSB?

The last bit (rightmost bit) of each color value is called **Least Significant Bit (LSB)**. Changing this bit changes the value very little, so the human eye cannot notice the difference.

Example (Red)

10110110 = 182

↓

10110111 = 183

Difference = 1
(Not visible to eyes)

HOW LSB STEGANOGRAPHY WORKS (EXAMPLE)

1. Secret Message

Let's hide the message: **HI**

Convert to ASCII binary

H = 01001000

I = 01001001

Message bits stream

0100100001001001

➡

2. Cover Image Pixels (Before)

Pixel No.	Red (8 bits)	Green (8 bits)	Blue (8 bits)
1	11010110	10110011	01010110
2	11100010	10011001	00101101
3	01101001	11001100	10100111
4	10011100	01100110	11110001
...	...	...	...

We will replace the LSB (last bit) of each color with message bits one by one.

↓

3. Embedding Process (Replace LSB with message bits)

Message bits stream

0 1 0 0 1 0 0 0

0 1 0 0 1 0 0 1

...

Pixel No.	Red (8 bits)	Green (8 bits)	Blue (8 bits)	Bits Used
1	11010110 → 11010110 0	10110011 → 10110011 1	01010110 → 01010110 0	0 1 0
2	11100010 → 11100011 1	10011001 → 10011000 0	00101101 → 00101100 0	0 1 0
3	01101001 → 01101000 0	11001100 → 11001100 0	10100111 → 10100110 0	0 0 0
4	10011100 → 10011101 1	01100110 → 01100110 0	11110001 → 11110001 1	1 0 1
...	...	...	...	...

4. Stego Image Pixels (After)

Only LSBs changed. Image looks the same!



Cover Image

=



Stego Image

5. Extraction Process

Read the LSB of each color in the same order.

Extracted bits

0 1 0 0 1 0 0 0

0 1 0 0 1 0 0 1

...

➡

Binary to ASCII

01001000 01001001

↓ ↓

H I

Recovered Message: HI

6. Reconstruct Message

Group every 8 bits and convert from binary to ASCII.

Key Points

- We change only the Least Significant Bit (LSB).
- Change is too small to be visible.
- More bits we embed, higher the chance of detection.
- Simple but very effective technique.

Pros and Cons of LSB

Pros

- Easy to implement
- High capacity
- Fast
- Good image quality

Cons

- Not very secure
- Can be destroyed by compression, resizing, cropping, or editing

2: Objectives

The objectives of Img-Crypt are:

- Hide secret messages inside images.
- Retrieve hidden messages from images.
- Protect hidden messages using password-based encryption.
- Provide a beginner-friendly interface.
- Support Windows and Kali Linux.
- Serve educational and research purposes.

3: Features

Message Encoding

- Hide text inside PNG or BMP images.
- Manual text input.
- Load text from a TXT file.
- Image size validation.
- Output image generation.

Message Decoding

- Extract hidden text.
- Detect encrypted messages automatically.
- Password verification.
- Save long messages to a text file.

Security

- Optional password protection.
- Fernet symmetric encryption.
- Base64 key generation.

User Experience

- Colored terminal interface.
- Animated disclaimer.
- KeyboardInterrupt handling.
- Error handling.
- Cross-platform compatibility.

4: Advantages

- Easy to use.
- Interactive.
- Password protection.
- Open source.
- Lightweight.
- Cross-platform.
- Educational.
- Beginner friendly.

5: Limitations

- Only text messages.
- No audio steganography.
- No video steganography.
- No PDF/document hiding.
- Single-image support.
- Uses only LSB.
- No metadata hiding.
- No compression.
- No steganalysis detection.